

Ayaan Retiwala

India | Remote

+91 92750 41139 | aretiwala@gmail.com | github.com/lolwierd | linkedin.com/in/lolwierd | lolwierd.com

Platform Engineer

I built the core of Excloud's cloud platform from scratch: compute, block storage, managed Kubernetes, networking, identity, and the SDK, CLI, console, and Terraform provider people use to drive it. The hard part of platform work is almost never the happy path, and that is the half I am good at.

EXPERIENCE

Excloud, Platform Engineer Remote

Jun 2023 to Aug 2026

Public cloud with **4,000+ accounts**, **380 active VMs**, and **140 managed Postgres clusters** at handover; built and operated the core platform alongside the engineering lead and handled escalations directly.

- Built the Go compute control plane over QEMU/KVM: VM lifecycle APIs, rescue boot, volume attach and reap, subnet operations, bare-metal host setup, and reconciliation after failed transitions.
- Designed and built the block storage engine on SPDK and NVMe-oF, plus the VFIO-user path into QEMU guests. A reconciler converges database intent against live bdev, lvol, and RAID state, so a crash mid-provision recovers instead of orphaning volumes.
- Shipped managed Kubernetes end to end: control-plane and worker bootstrap, OIDC and JWKS, short-lived join credentials, Cilium and CoreDNS, a CSI driver with attach, mount, resize, and failover, and a Karpenter cloud provider.
- Wrote the shared Go layer the services are built on: an HTTP framework deriving OpenAPI 3.1 from Go types by reflection, a typed error taxonomy, domain types, and OpenTelemetry middleware, used by **20+ internal services**.
- Traced a borrowed-header lifetime bug through the HTTP/2 and HPACK layer of the Kubernetes API transport and fixed it without losing OpenTelemetry trace propagation.
- Built the identity and account plane (IMDS tokens, Valkey-backed caching, IAM authorization, encrypted secrets) and the S3-compatible object storage API with multipart uploads, presigned URLs, and SigV4 checksums.
- Orchestrated managed Postgres and Kafka/Streams on the compute and DNS control planes: node lifecycle, replicas, broker provisioning, topics, users, and ACLs.
- Built the API-to-client pipeline from OpenAPI 3.1 across eight services, feeding the exc SDK and a **20-group CLI** plus a generated Terraform provider with **30 resources** and **13 data sources**.
- Built the Next.js console for compute, buckets, Streams/Kafka, secrets, billing, and security groups, with an authenticated instance terminal over Xterm and WebSockets.
- Wrote an AF_PACKET ARP/NDP proxy with BPF filtering, VLAN handling, rate limits, GARP, and IPv6 DAD, plus authoritative DNS with AXFR, IXFR, TSIG, and weighted records.
- Rebuilt the live-disk migration tool around resumable sender and receiver sessions with checkpointing, pinned per-session TLS, heartbeats, and idempotent thaw recovery.
- Also shipped Firecracker-based ephemeral GitHub Actions runners on owned hardware, usage-based billing, the CLI release and operator surface, and the Astro documentation site.

PROJECTS

DBConsole, a Postgres workbench for web and macOS

Electron, Vite, Next.js, PostgreSQL

19 releases

- Connection management, query tabs, SQL safety checks before destructive statements run, server-side result streaming, schema graphs, query history, and encrypted named-query sync with conflict resolution.

CBManager, a native macOS clipboard manager

Swift, AppKit, SQLite, Vision

v1.0.1

- Menu-bar app with global shortcut and paste-back, OCR over copied images, fuzzy and semantic search, bounded caching, SQLite WAL writes, **162 test methods**, CI, and DMG packaging.

Rig, a dispatch console for the pi coding agent

React, Vite, Fastify, WebSockets

self-hosted

- Mobile-first session board with live tool-call streaming, model and thinking-level selection, resume and stop controls, Tailscale deployment, and Telegram or REST operator access.

Tachyon, a manga-reader PWA

offline-first

PWA

- Offline reading, source integration, auto-scroll, AniList sync, and a test suite built around async state and race conditions.

EDUCATION

Sardar Vallabhbhai Patel Institute of Technology

B.E., Computer Engineering

2018 to 2022

Full-time GATE (Graduate Aptitude Test in Engineering) preparation

2022 to 2023

SKILLS

LANGUAGES	Go, C, TypeScript, Swift, Python
INFRASTRUCTURE	Kubernetes, QEMU/KVM, Firecracker, SPDK, NVMe-oF, VFIO-user, CSI, Karpenter, Linux, systemd
DATA AND TELEMETRY	PostgreSQL, pgx, SQLite, Valkey, ClickHouse, Kafka/KRaft, OpenTelemetry
DEVELOPER PLATFORM	OpenAPI 3.1, code generation, Terraform Plugin Framework, Cobra, GitHub Actions
PRODUCT	Next.js, React, Electron, Vite, Tailwind CSS, Fastify, WebSockets, Xterm, Astro
NETWORKING	DNS, AXFR, IXFR, TSIG, ARP/NDP, IPv4/IPv6, HTTP/2, TLS